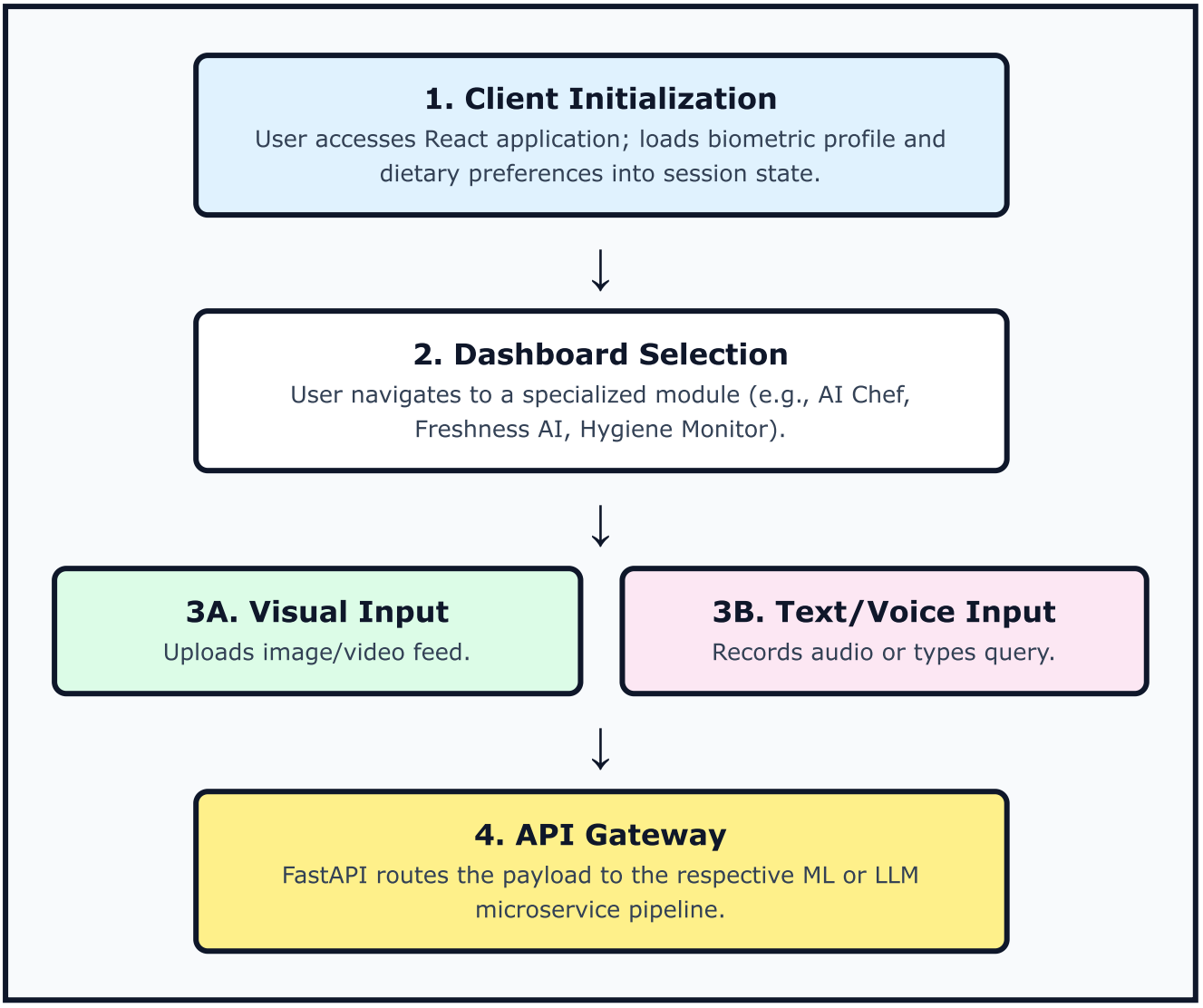
5. PROJECT WORKFLOW

The FoodVerse AI workflow is designed around a decoupled, event-driven architecture. User actions on the frontend trigger specific API pipelines on the backend, routing data either to the PyTorch vision models or the Groq-powered LLM engine depending on the active module.

Phase 1: High-Level Application Routing



Phase 2: Vision Processing Pipeline

When a user uploads an image for Food Recognition, Freshness Detection, or Hygiene Monitoring, the data strictly follows the visual processing lifecycle to minimize latency and ensure privacy.

1. Image Ingestion (FastAPI)

Multipart-form data is received. The API validates file type, limits payload size (Max 12MB), and stores the file temporarily in memory.

↓

2. OpenCV Preprocessing

The image is resized to standard dimensions (e.g., 224x224), normalized, and converted into a tensor format suitable for the neural network.

↓

3. PyTorch Model Inference

The tensor is passed through the respective trained vision model (e.g., YOLO for hygiene bounding boxes, or ResNet for freshness classification).

↓

4. Post-Processing & Formatting

Raw model logits are converted into confidence scores and mapped to human-readable labels (e.g., "Spoiled Apple: 92% Confidence").

↓

5. Frontend State Update

FastAPI returns a structured JSON response. React parses the JSON and immediately updates the UI dashboard with the metrics.

Phase 3: Generative AI Workflow

For cognitive tasks like the Smart AI Chef and Personalized Meal Planner, the workflow aggregates contextual data and utilizes prompt engineering before querying the LLM engine.

